Comparative Study of 30 Research Sources

Generative AI-Based Cloud Removal and Reconstruction for LISS-IV Satellite Imagery

Prepared as a literature-review companion document | IEEE-style numbered references | 9 October 2026

Important scope note. This review includes verified IEEE-published papers where publisher metadata was available, plus closely related peer-reviewed papers and foundational methods needed to understand the architectures. It is not accurate to label all 30 sources as IEEE-published. The links point to DOI, IEEE Xplore, proceedings, or dataset/project pages. Several search-only entries are explicitly marked as discovery links and should be replaced by the final publisher record before formal submission. Confirm authors, volume, pages, and DOI in IEEE Xplore / Crossref before submitting.

1. Executive summary

Cloud removal has evolved from interpolation and multitemporal compositing toward CNNs, adversarial image translation, multimodal SAR-optical fusion, attention-based networks, and conditional diffusion. The main trade-off is between computational efficiency, visual plausibility, spectral fidelity, and dependence on auxiliary data. For LISS-IV, sensor-specific adaptation and geospatial consistency are as important as model choice. A credible project should compare simple baselines first and clearly separate measured pixels from generated pixels.

2. Comparative matrix

#	Source / method	Family	Data / supervision	Main value	Caution for LISS-IV
1	Zheng et al. [1]	U-Net + GAN	Single optical image; supervised	Two-stage segmentation/restoration	GAN may hallucinate; adapt bands/resolution
2	Sui et al. [2]	Diffusion enhancement	Remote-sensing imagery	Fine detail and perceptual quality	High compute; test radiometry
3	DiffCR [3]	Conditional diffusion	Optical benchmarks	Efficient conditioning design	Benchmark transfer is not guaranteed
4	SEN12MS-CR-TS [4]	Dataset + 3-D CNN / sequence translation	Multitemporal optical + SAR	Multimodal temporal benchmark	Sentinel data differ from LISS-IV
5	DBD-CR [5]	Diffusion + residual correction	Three cloud-removal datasets	Corrects missing texture residuals	Check uncertainty and latency
6	SAR + deep learning [6]	SAR-guided inpainting	Optical + SAR	Uses all-weather structural cues	SAR does not provide optical colour
7	Coupled tensor factorization [7]	Tensor factorization	Multitemporal images	Robustness to imperfect masks	Less semantic than deep models
8	TSMCF [8]	Transformer cross-attention	SAR + multispectral optical	Adaptive cross-modal fusion	Requires precise co-registration
9	Hybrid 3-D/2-D fusion [9]	Multisource deep fusion	Sentinel-1/2	Fuses spatial and spectral cues	Domain gap and compute
10	Cloud Meets Diffusion [10]	Progressive diffusion	Optical imagery	Progressive removal without fixed thin/thick classes	Newer method; verify full metadata
11	LFDBP-Net [11]	Unsupervised disentanglement + prompting	Unpaired data	Reduces paired-data dependency	Complex optimization; validate generalization
12	Transfer-learning GAN [12]	Transfer learning + GAN	Satellite imagery	Potentially reduces data burden	Pretraining domain may differ
13	Efficient SAR-optical network [13]	Feature pyramid fusion	SAR + optical	Efficiency-oriented reconstruction	Use exact final DOI record
14	GAN survey [14]	Survey	Satellite imagery literature	Taxonomy and evaluation context	Broad survey, not a direct baseline
15	Cloud-deterministic diffusion [15]	Progressive diffusion	Optical imagery	Cloud-aware generation direction	Very recent; independently verify details
16	Meraner et al. [16]	Residual CNN + SAR-optical	Sentinel-1/2 real pairs	Cloud-adaptive loss and sensor fusion	Sensor and band adaptation needed
17	Multitemporal SAR/optical [17]	CNN fusion	Sentinel-1/2 time series	Accounts for ground changes	Date alignment and temporal availability
18	Spatiotemporal interactive learning [18]	SAR-optical temporal fusion	Multitemporal imagery	Joint temporal/modal features	Need reliable image registration
19	Grich et al. [19]	Gap-filling comparison	Sentinel-2	Compares spatial/temporal/hybrid approaches	New publication; confirm dataset settings
20	SEN12MS resources [20]	Dataset / fusion	Sentinel-1/2	Useful multisensor pretraining resource	Not a LISS-IV ground-truth dataset
21	CR-TS implementation [21]	Dataset/code resource	Multitemporal SAR/optical	Reproducible baseline access	Check version, license, and split
22	Deep-learning cloud-removal discovery [22]	CNN literature	Varies	Broad method discovery	Search link; replace with exact paper citation
23	RICE dataset discovery [23]	Dataset benchmark	Optical image pairs	Useful benchmark exploration	Verify paper version and license
24	GAN cloud-removal discovery [24]	GAN literature	Varies	Find related adversarial baselines	Search link; exact paper must be selected
25	CycleGAN [25]	Unpaired image translation	Unpaired domains	Useful when paired images scarce	Unpaired translation may change semantics
26	GANs [26]	Generative adversarial learning	General method	Adversarial objective foundation	Not satellite-specific
27	U-Net [27]	Encoder-decoder CNN	Supervised segmentation	Strong compact baseline	Not a cloud-removal paper originally

#	Source / method	Family	Data / supervision	Main value	Caution for LISS-IV
28	DDPM [28]	Diffusion foundation	General image generation	Principled denoising generation	Sampling latency can be high
29	Transformer [29]	Attention architecture	General sequence modeling	Long-range feature interactions	Memory cost for high-resolution patches
30	SEN12MS [30]	Dataset	Sentinel-1/2	Multimodal pretraining/data fusion	Band/resolution mismatch with LISS-IV

3. Cross-method comparison

Criterion	Interpolation / temporal	CNN / GAN	SAR fusion / attention	Diffusion
Training data	Low for interpolation; temporal images needed	Usually paired cloudy/clear data	Aligned SAR + optical; often supervised	Large/varied training data preferred
Compute	Low	Low–medium	Medium–high	Often high at inference
Visual detail	Low for large gaps	Good; GAN can sharpen	Good where SAR gives structure	Often strong texture and global plausibility
Spectral fidelity	Depends on reference	Must be explicitly constrained	Needs modality-aware losses	Needs explicit radiometric constraints
Temporal change	May fail if reference stale	Single-image model may hallucinate	Can use current SAR plus temporal optical	Depends on conditioning
Main failure	Blur / stale data	Hallucination / oversmoothing	Misregistration / modality gap	Slow inference / plausible but wrong detail
Best role	Baseline / gap fill	Practical primary model	Auxiliary branch when SAR available	Advanced comparison after baseline

4. Recommended architecture for the five-member team

Stage 1 — reproducible baseline: implement cloud mask handling, patch-based preprocessing, and U-Net reconstruction with L1 + SSIM losses. Preserve clear pixels by mask-aware blending. This yields an interpretable benchmark and reduces the risk of starting with a difficult diffusion system.

Stage 2 — generative enhancement: add a conditional discriminator and a conservative adversarial loss. Compare sharpness and spectral metrics against the U-Net baseline. Reject settings that improve visual appearance while worsening SAM or downstream classification.

Stage 3 — auxiliary information: if properly aligned SAR or near-date optical scenes are available, add a separate encoder and attention-based fusion. If not, do not claim SAR-assisted reconstruction. Report the data actually used.

Stage 4 — diffusion extension: evaluate a conditional diffusion model only if training data and GPU time are sufficient. Compare latency, PSNR, SSIM, RMSE, SAM, and cloud-region-only scores on the same test scenes.

5. Evaluation protocol

Use scene-level train/validation/test splits; synthetic masks for paired metrics and real cloudy scenes for qualitative generalization; separate thin cloud, thick cloud, shadow, water, vegetation, and urban strata; and report both cloud-only and whole-image scores. Recommended metrics are PSNR, SSIM, RMSE, SAM, inference latency, and uncertainty calibration. Add a downstream task such as land-cover classification to test whether reconstructed imagery preserves analytical usefulness. Never report values copied from a paper as the performance of this project.

6. Team division of work

Member	Suggested responsibility	Deliverables
Member 1	Literature review and data sourcing	Reference verification, LISS-IV/Sentinel data manifest, data permissions
Member 2	Preprocessing and cloud masks	Radiometric scaling, co-registration, tiling, mask generation
Member 3	Baseline model	U-Net training, loss functions, checkpoint and ablation runs
Member 4	Generative / fusion model	Conditional GAN or SAR/temporal branch; only if data are available
Member 5	Evaluation, UI, report integration	Metrics, visual comparisons, GeoTIFF export, documentation

7. Reference list with access links

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